HCP TARGETING &

BRAND PERFORMANCE ANALYTICS

Diabetes Portfolio — Case Study

12-Week Upskilling Project | Commercial Analytics

CMS Medicare Part D 2021-2022 | ~227K HCPs | 83M+ Rx Rows

Python | PostgreSQL | Streamlit | openpyxl | sklearn

Executive Summary

This case study documents a 12-week analytics engineering project building a commercial analytics platform for a diabetes drug portfolio. The platform transforms publicly available Medicare claims data into actionable intelligence across five sales personas — enabling field reps, area managers, regional managers, and national leadership to prioritise HCP engagement, identify territory gaps, and model the financial impact of targeting decisions.

Dimension	Detail
Dataset	CMS Medicare Part D 2021-2022 | ~227,000 HCPs | 83M+ prescription rows
Additional	CMS Open Payments 2022 — KOL advisory & speaker fees by NPI
Stack	Python 3.11 | Streamlit | PostgreSQL (Neon serverless) | SQLAlchemy | scikit-learn
Deployment	Streamlit Community Cloud — live public URL
Personas	Sales Rep | Area Manager | Regional Manager | Head of Sales
Key Outputs	HCP priority scoring (ML + rule) | Territory ROI model | Drug trend analysis | White space mapping
Duration	12 weeks — structured upskilling exercise

> *The platform surfaces insights that took hours of manual spreadsheet work in under 3 seconds of SQL query time — and exposes them through a role-personalised UI that requires no SQL knowledge from the end user.*

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1. Problem Statement

Commercial pharma field teams face a persistent prioritisation problem. With hundreds of HCPs in a territory and limited rep capacity, the question is always: who do I call first, what do I say, and when do I call them back? Without data-driven guidance:

- High-prescribing HCPs get over-called while high-potential new entrants go undiscovered
- Area and regional managers cannot spot territory coverage gaps without running manual reports
- Brand intelligence — which drug classes are growing, where prescribers are switching — is disconnected from field planning
- KOL relationships are managed reactively rather than proactively

This project builds the infrastructure to answer all four of those questions through a live, interactive platform that updates with each data refresh.

2. Data Sources & Architecture

2.1 Data Sources

Source	Volume	Key Fields Used
CMS Medicare Part D 2022	83M+ Rx rows → HCP level	fills, drug_cost, drug_name, specialty, state
CMS Medicare Part D 2021	Aggregated to HCP level	fills_prior_year — YoY growth calculation
CMS Open Payments 2022	~12M payment records	NPI, total_payment_usd, opinion_leader_payments
NPPES NPI Registry	~8M provider records	npi, specialty, state, city
CDC T2D Prevalence	State-level estimates	State T2D prevalence rate — white space analysis

2.2 Derived Analytics Fields

Field	Logic	Use
targeting_score	0.5×(vol_decile/10) + 0.3×(growth_decile/10) + 0.2×(kol_flag)	Primary sort key — rule-based baseline
ml_targeting_score	GBR trained on 7 features to predict fills_2022; normalised 0-1	ML-derived priority score (live in app)
segment	CASE: vol+growth>=8 → HV; growth>=8 → Growth; vol>=5 → Maint; else Deprioritise	Cadence and messaging strategy
yoy_growth_pct	(fills_2022 - fills_2021) / fills_2021 * 100	Momentum signal
priority_label	P1 if score>=0.72; P2 if>=0.50; else P3	Rep-facing priority triage

2.3 Technology Stack

Layer	Technology	Purpose
Database	Neon serverless PostgreSQL + pgBouncer	Persistent HCP + drug trend data; cold-start tolerant
ORM / Query	SQLAlchemy + pandas	Connection pooling, DataFrame transforms
Application	Python 3.11 + Streamlit 1.35	Multi-persona interactive dashboard

Layer	Technology	Purpose
ML	scikit-learn GradientBoostingRegressor	Live HCP targeting score trained on CMS data
Compute	Streamlit Community Cloud	Zero-infra deployment; live public URL
Modelling	openpyxl + Excel formula engine	Territory ROI calculator (Week 10 deliverable)
Version Ctrl	Git + GitHub	Portfolio-ready public repository

3. What Was Built

3.1 Sales Rep View

The rep view is isolated from all other tabs via st.stop() — preventing senior personas from loading rep-specific query logic. Key features:

• Priority 1/2/3 HCP cards colour-coded by targeting score threshold (0.72 / 0.50), showing both ML and rule scores side-by-side
• Pre-call brief generator: contextual talking points derived from segment, KOL status, growth trajectory, and fill volume
• Call due status with traffic-light colour: Overdue / Due Soon / OK — per HCP segment cadence rule
• One-click 'Mark as Called' simulation updating session state — demonstrates CRM write-back pattern
• Next Best Action: rule-based engine (6 action types) surfaced per HCP

3.2 Area Manager — 4 Tabs

Tab	Content
My Team	Rep scorecards (5 simulated reps via NPI modulo) with P1 HCP count, calls logged, overdue alerts, grouped bar chart
Territory Gaps	Uncovered HCPs table: overdue or dark 30+ days. KPIs: total uncovered, high-value uncovered, estimated revenue at risk
Strategic Accounts	KOL accounts + High Value HCPs with >-3% YoY decline. Expandable pre-call briefs per HCP
Territory Health	KPIs: P1 coverage, avg targeting score, churn risk rate, momentum. Segment donut + at-risk table with conversion targets

3.3 Regional Manager — 4 Tabs

Tab	Content
My AMs	Area Manager scorecards (one per state: TX, AZ, NM, OK in Southwest region). Grouped bar chart by segment per state
Regional Trends	YoY fill change bar chart by state. Segment migration cards: rising HCPs (HV + >10% YoY) vs declining (Maint + <-5% YoY)
KOL Intelligence	KOLs split: engagement gap (advisory payments but declining Rx — competitive threat) vs healthy relationships
White Space	Scatter: T2D prevalence vs HCP coverage % by state. Bubble sized by HCP count, coloured by opportunity level

3.4 Head of Sales — 4 Tabs

Tab	Content
National Scorecard	4 national KPIs + region performance cards (RAG rating) + grouped bar by region
Franchise Accounts	Top 50 HCPs nationally by fills_2022. Trend signal, KOL flag, days since last contact. At-risk alert for YoY decliners
Brand Intelligence	Drug trend analysis from v_drug_trends view: prescriber base and fill volume over time by drug class
Strategic Priorities	Left: HV declining HCPs with estimated revenue at risk ($). Right: new entrant detection and underserved states

4. Key Analytical Insights Surfaced

Insight	Signal	Commercial Implication
KOL engagement gap	KOL receiving advisory payments but with YoY Rx decline	Competitor may have made a more compelling advisory offer. Requires executive intervention — not a field rep visit
New entrant detection	growth_decile >= 8, fills_2021 <= 50, fills_2022 >= 20	Rapidly growing low-base prescribers — earliest stage to establish brand preference before competitors win them
Segment migration	Rising: HV + YoY > 10%. Declining: Maintenance + YoY < -5%	Leading indicator of next-quarter reclassification — enables proactive intervention before official re-ranking
White space by state	T2D prevalence high but HCP coverage rate low	Market opportunity not proportional to rep deployment. Signals need for territory rebalancing or remote detailing
Cadence miss rate	HCPs where days_dark > CALL_CADENCE_DAYS by segment	Systematic coverage gaps detectable at territory level — an Area Manager metric, not just a rep coaching point

Most commercially impactful insight class: the KOL engagement gap. A KOL with advisory payments but declining Rx is the strongest competitive threat signal in the dataset. It does not appear in standard CRM reporting — it requires joining claims data with Open Payments, which this platform does automatically.

5. ML Scoring — GBR Implementation

A Gradient Boosting Regressor is trained on 7 CMS features at app startup (cached per session, trains once in ~2 seconds). Target is fills_2022 — actual prescribing volume — predicted from prior-year data and contextual signals. This mirrors the production use case: predict next year's prescribing from current-year features.

Feature	Source	What It Captures
fills_2021	CMS Part D 2021	Prior-year Rx volume — strongest predictor of current volume
volume_decile	Derived (NTILE)	Specialty-normalised volume rank 1-10
growth_decile	Derived (NTILE)	Specialty-normalised growth rank 1-10
yoy_growth_pct	Derived	Year-on-year change — momentum direction
is_kol	CMS Open Payments	KOL status binary — advisory/speaker fee received
t2d_prev	CDC estimates	State T2D prevalence — market size proxy
specialty_enc	NPPES (label enc.)	Specialty group — controls for specialty mix effects

> *The ML score surfaces non-linear patterns the rule formula misses: a KOL with declining fills is penalised more; a high-growth low-base doctor is promoted more aggressively. Both Rule and ML scores are shown side-by-side on every HCP card so reps see the signal from both approaches.*

6. Territory ROI Calculator

A standalone five-sheet Excel model built in openpyxl quantifies the commercial impact of targeting decisions. All calculations use Excel formulas referencing an Inputs sheet — no hardcoded values — so the model recalculates dynamically when assumptions change.

Sheet	Purpose	Key Calculations
Cover	Navigation index with hyperlinks	—
Inputs	All assumptions — blue = input cells	Call cost $180, Rev/fill $45, base fills 120, cadence days by segment, Rx lift rates, HCP counts
Segment ROI	Annual ROI by segment	Calls/year, call cost, revenue lift, net return, ROI% — all Excel formulas cross-referencing Inputs
Territory Compare	Side-by-side 4 territory configurations	Cross-sheet references (green) pulling Inputs values — enables instant scenario comparison
Scenario Analysis	Best / Base / Worst case sensitivity	=IF(scenario='Best', Inputs!C6*1.2, ...) — dynamic per cell
Priority Matrix	2x2 ROI vs churn risk heatmap	Conditional logic: High ROI + Low Risk = Priority 1 action

Metric	Value
Total Excel formulas	102
Formula errors on delivery	0
Colour convention	Blue = inputs | Black = formulas | Green = cross-sheet links
Verification	LibreOffice headless recalc.py + full cell scan for #REF! / #DIV/0! / #VALUE!

7. Production Architecture

The prototype uses CMS public data and a shared serverless database. A production deployment would differ in three key dimensions:

Dimension	Prototype	Production
Data	CMS Medicare Part D — public, lagged 12+ months	Specialty Pharma SHS / IQVIA TRx — weekly refresh, actual brand-level prescribing
Call logs	Simulated via NPI modulo + datetime seeding	Veeva CRM or Salesforce Health Cloud — real call dates and outcomes
Database	Neon serverless PostgreSQL (shared)	Dedicated cloud PostgreSQL / Snowflake with role-based access control
Auth	Streamlit persona selector dropdown	SSO via Okta + Salesforce/Veeva identity — persona locked by user role
KOL data	CMS Open Payments (all specialties, all pharma)	Brand-specific advisory board and speaker bureau data
Refresh	Annual (CMS data cycle)	Weekly (SHS/IQVIA) or daily (CRM events)
ML	GBR on CMS features — self-supervised on public data	Supervised churn classifier + NBA multi-label model on CRM call history

Architecture principle: swapping data sources requires changing only the ETL layer and connection string — not the application logic. All persona views, scoring functions, and KPI calculations reference column names that remain constant across CMS and commercial data sources.

8. Skills Demonstrated

Domain	Specific Skills
Data Engineering	ETL pipeline design | SQL aggregation (window functions, CTEs, VIEWs) | PostgreSQL + SQLAlchemy | Neon serverless cold-start handling
Analytics	HCP segmentation | YoY growth analysis | KOL engagement gap detection | White space via T2D prevalence overlay | Commercial ROI modelling
Machine Learning	GBR implementation (scikit-learn) | Feature engineering on CMS data | MinMaxScaler normalisation | Feature importance extraction | Caching strategy
Software Engineering	Multi-persona Streamlit app | st.stop() isolation | Session state | Deterministic NPI-seeded simulation | GitHub + CI deployment
Commercial Acumen	Call cadence strategy by segment | Revenue at risk quantification | New entrant detection | KOL relationship risk identification
Financial Modelling	Territory ROI calculator | Industry colour convention | Formula-only Excel model | Scenario and sensitivity analysis

9. What's Next

- Connect real commercial TRx feed (IQVIA / SHS) to replace CMS lagged data
- Train supervised churn classifier once CRM call history (2+ years) is available
- Deploy XGBoost churn model with Salesforce Health Cloud write-back
- Add Veeva CRM call log integration to replace simulated last-call dates
- Implement SSO authentication (Okta / Salesforce Identity) to replace dropdown persona selector
- Build dynamic cadence engine (Weibull survival analysis) on actual interaction data

This document is a portfolio / upskilling exercise artefact. All data used is publicly available from the US Centers for Medicare & Medicaid Services. No proprietary, confidential, or client data has been used.